



THE KING'S SCHOOL

2007
Higher School Certificate
Trial Examination

Mathematics Extension 1

General Instructions

- Reading time – 5 minutes
- Working time – 2 hours
- Write using black or blue pen
- Board-approved calculators may be used
- A table of standard integrals is provided at the back of this paper
- All necessary working should be shown in every question

Total marks – 84

- Attempt Questions 1-7
- All questions are of equal value

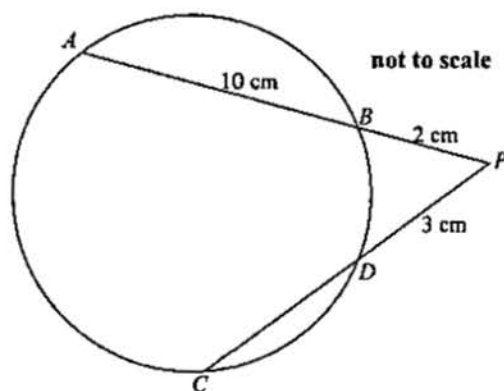
Total marks – 84
Attempt Questions 1-7
All questions are of equal value

Answer each question in a SEPARATE writing booklet. Extra writing booklets are available.

Question 1 (12 marks) Use a SEPARATE writing booklet.

Marks

(a)



The diagram shows the chord AB and CD intersecting externally at P .

If $AB = 10$ cm, $BP = 2$ cm and $DP = 3$ cm find the length of the chord CD .

2

(b) $P(3, 5)$ divides the interval AB externally in the ratio $m : n$.

If $A = (-6, -13)$ and $B = (-1, -3)$, find the ratio $m : n$.

2

(c) Differentiate $\frac{x}{1 + e^{2x}}$

2

(d) Find $\int 2\sin^2 x \, dx$

2

(e) Solve $\frac{x^2 + 2}{x + 2} < 1$

3

(f) Find $\lim_{x \rightarrow 0} \frac{x^2}{\sin x}$

1

End of Question 1

(a) Find the acute angle between the lines $y = \frac{2}{3}x + 1$ and $y = -\frac{1}{5}x - 1$ 2

(b) Use the substitution $x = u^2, u > 0$, to evaluate $\int_1^4 \frac{dx}{\sqrt{x}(1 + \sqrt{x})}$ 4

(c) The cubic equation $x^3 + ax^2 + bx + c = 0$ has roots $-\alpha, \alpha$ and $\frac{1}{\alpha}$

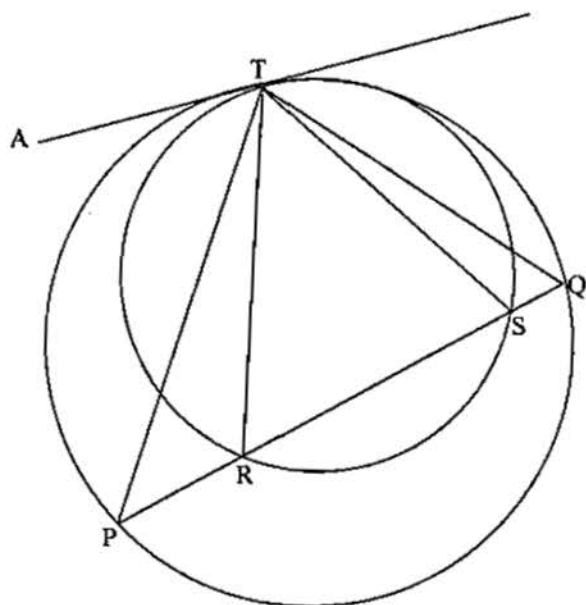
(i) Show that $ac = -1$ 2

(ii) Show that $b + c^2 = 0$ 2

(d) Find $\int_0^3 \frac{dx}{\sqrt{16 - x^2}}$. Give your answer correct to two decimal places. 2

End of Question 2

(a)



The diagram shows two circles touching internally at T. AT is the common tangent to the circles. The chord PQ of the outer circle meets the inner circle at R and S.

(i) Explain why $\angle ATP = \angle TQS$

1

(ii) Prove that $\angle PTR = \angle QTS$

3

(b) $f(x) = \tan^{-1} x - x + 0.3 = 0$ has a root near $x = 1.2$

Use one application of Newton's method to find a second approximation to this root.

Give your answer correct to 2 decimal places.

3

(c) A particle moves on the x axis in simple harmonic motion. Its velocity v at any time $t \geq 0$ is given by $v = 6 \sin 3t$.

Initially the particle is at rest at $x = -1$

(i) At what time is the particle next at rest?

1

(ii) Hence or otherwise give the period of the motion.

1

(iii) Find an expression for the displacement x in terms t .

2

(iv) State when the particle is first at the position $x = 1$

1

End of Question 3

(a) (i) Show that $\cos\left(\frac{\pi}{4} - \frac{\pi}{12}\right) - \cos\left(\frac{\pi}{4} + \frac{\pi}{12}\right) = 2 \sin \frac{\pi}{4} \sin \frac{\pi}{12}$ 1

(ii) Hence prove that $\frac{\cos \frac{\pi}{6} - \cos \frac{\pi}{3}}{\sin \frac{\pi}{6} + \sin \frac{\pi}{3}} = \tan \frac{\pi}{12}$ 2

(iii) Hence or otherwise show that $\tan \frac{\pi}{12} = 2 - \sqrt{3}$ 2

- (b) The surface area S of a sphere is increasing at a constant rate of $k \text{ cm}^2/\text{s}$. Prove that the volume V of the sphere is increasing at a rate proportional to the radius r at any time t .

$\left[S = 4\pi r^2, V = \frac{4}{3}\pi r^3 \right]$ 3

- (c) Consider the function $f(x) = \frac{2}{\pi} \cos^{-1}(1 - x)$

- (i) Find the domain of f . 2

- (ii) Sketch the function. 2

End of Question 4

- (a) Prove by mathematical induction for positive integers n that

$$(1^3 + 3 \cdot 1^5) + (2^3 + 3 \cdot 2^5) + \dots + (n^3 + 3n^5) = \frac{1}{2}n^3(n+1)^3$$

$$[\text{ You may assume } (n+2)^3 = n^3 + 6n^2 + 12n + 8]$$

3

- (b) Let $(1+x)^n = c_0 + c_1x + c_2x^2 + \dots + c_kx^k + \dots + c_nx^n$

(i) Show that $\frac{c_k}{c_{k-1}} = \frac{n-k+1}{k}$, $1 \leq k \leq n$

2

(ii) Deduce that $\frac{c_1}{c_0} + \frac{2c_2}{c_1} + \frac{3c_3}{c_2} + \dots + \frac{kc_k}{c_{k-1}} + \dots + \frac{nc_n}{c_{n-1}} = \frac{n}{2}(n+1)$

2

- (c) Let $f(x) = x^3 + (A+2)x^2 + Ax - 1$

(i) Show that $x+1$ is a factor of $f(x)$ for all values of A

1

(ii) If $(x+1)^2$ is a factor of $f(x)$ find A

2

(iii) If $x^2 - Ax - 1$ is a factor of $f(x)$ find A .

2

End of Question 5

(a) Let $f(x) = \ln\left(\tan\frac{x}{2}\right)$, $0 < x < \pi$

(i) Show that $f'(x) = \operatorname{cosec} x$ 3

(ii) Sketch the curve showing the x intercept and any asymptotes. 2

(iii) Find the inverse function $y = f^{-1}(x)$ 2

(b) A particle moves on the x axis with its velocity v given by $\frac{dx}{dt} = 20 - x$

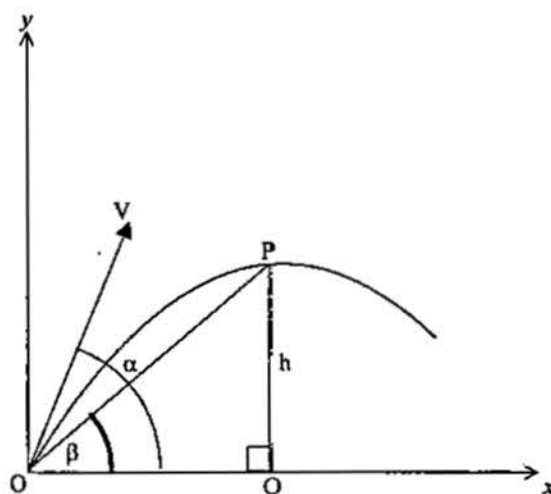
Initially the particle is at $x = 19$

(i) Prove that the acceleration is given by $\ddot{x} = -v$ 2

(ii) Hence express v as a function of t . 3

End of Question 6

(a)



A particle is projected from O with velocity V at an angle of α to the horizontal.

At time t the equations of motion are:

$$\dot{x} = V \cos \alpha$$

$$\dot{y} = -10t + V \sin \alpha$$

$$x = (V \cos \alpha) t$$

$$y = -5t^2 + (V \sin \alpha) t$$

[DO NOT PROVE THESE]

After time T the particle reaches its greatest height $h = PQ$ where $\angle POQ = \beta$

(i) Show that $T = \frac{V \sin \alpha}{10}$ 1

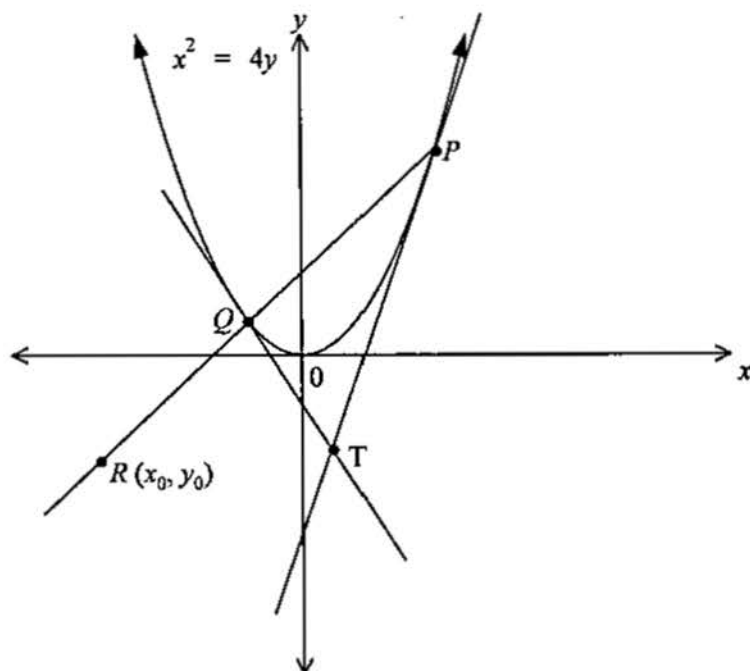
(ii) Deduce that $\tan \beta = \frac{1}{2} \tan \alpha$ 2

(iii) Show that $20h = V^2 \sin^2 \alpha$ 1

(iv) Deduce that $V^2 = 5h(4 + \cot^2 \beta)$ 2

Question 7 continues on next page

(b)



The diagram shows a variable chord PQ of the parabola $x^2 = 4y$ passing through the point $R(x_0, y_0)$. The tangents at P and Q meet at T .

The equation of the tangent at a point (x_1, y_1) on $x^2 = 4y$ is $x_1 x = 2(y + y_1)$.

The equation of the chord of contact from $R(x_0, y_0)$ to $x^2 = 4y$ is $x_0 x = 2(y + y_0)$.

[DO NOT PROVE THESE]

- (i) Let $P = (2p, p^2)$

Show that the equation of the tangent at P is $px = y + p^2$ 1

- (ii) Show that the tangents at $P(2p, p^2)$ and $Q(2q, q^2)$ meet at $T(p + q, pq)$ 2

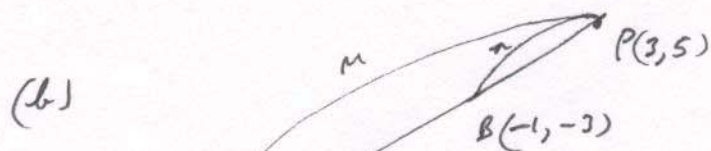
- (iii) Show that the equation of the chord PQ is $(p + q)x = 2(y + pq)$ 1

- (iv) Hence prove that the locus of T lies on the line containing the chord of contact from $R(x_0, y_0)$ 2

End of Examination

Question 1(a) Let $CD = x$

$$\text{Then } 3(x+3) = 12 \times 2 \Rightarrow x = 5$$

 $\therefore$ considering x coordinates

$$m : n = 9 : 4$$

(c)
$$\frac{(1+e^{2x}) - x(2e^{2x})}{(1+e^{2x})^2} \quad \text{will do}$$

$$= \frac{1 + e^{2x} - 2xe^{2x}}{(1+e^{2x})^2}$$

(d)
$$= \int 1 - \cos 2x \, dx = x - \frac{\sin 2x}{2} (+ C)$$

(e) (Alternatives abound)

For $x+2 > 0$ i.e. $x > -2$ we have $x^2+2 < x+2$

$$\Rightarrow x^2 - x < 0$$

$$x(x-1) < 0 \quad \therefore 0 < x < 1$$

 $\therefore 0 < x < 1$ is a solution as is $x < -2$

(f)
$$= \lim_{x \rightarrow 0} x \cdot \frac{x}{\sin x} = 0 \times 1 = 0$$

Question 2

(a) $m_1 = \frac{2}{3}, m_2 = -\frac{1}{5}$

$$\therefore \tan \alpha = \frac{\frac{2}{3} + \frac{1}{5}}{1 - \frac{2}{3} \cdot \frac{1}{5}} = \frac{10+3}{15-2} = 1 \quad \therefore \alpha = 45^\circ$$

(b) $x = u^2$

$x=1, u=1$

$x=4, u=2$

$$\frac{dx}{du} = 2u$$

$$\therefore I = \int_1^2 \frac{2u \, du}{u(1+u)} = 2 \int_1^2 \frac{1}{1+u} \, du$$

$$= 2 [\ln(1+u)]_1^2$$

$$= 2(\ln 3 - \ln 2) = 2 \ln 1.5$$

(c) (i) Sum of roots $= \frac{1}{2} = -a$

Product of roots $= -2 = -c$

$$\therefore ac = \frac{1}{2}(-2) = -1$$

(ii) $-2 \cdot 2 - 2 \cdot \frac{1}{2} + 2 \cdot \frac{1}{2} = -2^2 = b$

$$\therefore \text{From (i), } b + 2^2 = b + c^2 = 0$$

(d) $I = \left[\sin^{-1} \frac{x}{4} \right]_0^3 = \sin^{-1} \left(\frac{3}{4} \right) = 0.85, 2 \text{ d.p.}$

Question 3

(a)(i) alternate segment theorem in the outer circle

(ii) $\angle PTR = \angle TSR$, alt. seg. thm in inner circle

$\therefore \angle PTR = \angle QTS$ from (i) and exterior angle theorem in $\triangle QTS$.

[Sadly (for the marker), lots of alternatives]

(b) $f'(x) = \frac{1}{1+x^2} - 1$

$$\therefore x_1 = 1.2 - \frac{\tan^{-1} 1.2 - 1.2 + 0.3}{\frac{1}{1+1.2^2} - 1} = 1.16, 2 \text{ d.p.}$$

(c) (i) $v=0 \Rightarrow \sin 3t = 0$

$\therefore 3t = \pi$ for next at rest

i.e. $t = \frac{\pi}{3}$

(ii) From (i), period $= 2 \times \frac{\pi}{3} = \frac{2\pi}{3}$

(iii) $\frac{dx}{dt} = 6 \sin 3t \quad \therefore x = -\frac{6 \cos 3t}{3} + c$

$$\therefore -1 = -2 + c, \quad c = 1$$

$$\therefore x = 1 - 2 \cos 3t$$

$\Rightarrow$ particle oscillates about $x=1$

(iv) From (iii), at $x=1$ when $t = \frac{1}{2} \cdot \frac{\pi}{3} = \frac{\pi}{6}$

[for first time]

Question 4

$$(a) (i) LS = \cos \frac{\pi}{4} \cos \frac{\pi}{12} + \sin \frac{\pi}{4} \sin \frac{\pi}{12} - \left(\cos \frac{\pi}{4} \cos \frac{\pi}{12} - \sin \frac{\pi}{4} \sin \frac{\pi}{12} \right) \\ = 2 \sin \frac{\pi}{4} \sin \frac{\pi}{12}$$

$$(ii) \sin \frac{\pi}{6} + \sin \frac{\pi}{3} = \sin \left(\frac{\pi}{4} - \frac{\pi}{12} \right) + \sin \left(\frac{\pi}{4} + \frac{\pi}{12} \right) \\ = 2 \sin \frac{\pi}{4} \cos \frac{\pi}{12}$$

$$\therefore LS = \frac{2 \sin \frac{\pi}{4} \sin \frac{\pi}{12}}{2 \sin \frac{\pi}{4} \cos \frac{\pi}{12}} = \tan \frac{\pi}{12}$$

$$(iii) \text{ From (ii), } \tan \frac{\pi}{12} = \frac{\frac{\sqrt{3}}{2} - \frac{1}{2}}{\frac{1}{2} + \frac{\sqrt{3}}{2}} = \frac{\sqrt{3}-1}{\sqrt{3}+1}$$

$$= \frac{(\sqrt{3}-1)^2}{3-1} = \frac{4-2\sqrt{3}}{2} = 2-\sqrt{3}$$

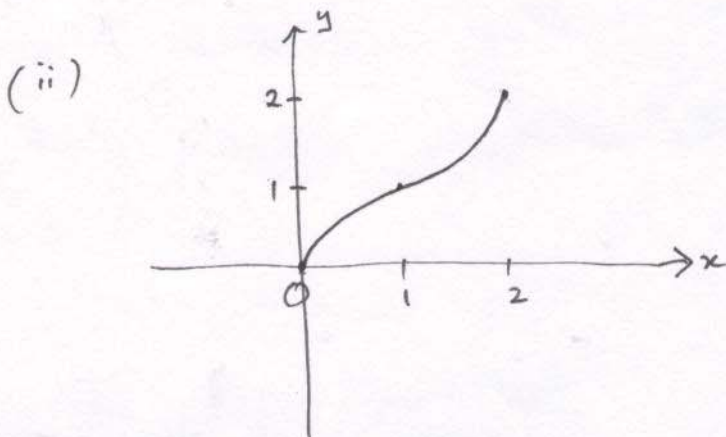
$$(b) \frac{dS}{dt} = k \quad \& \quad \frac{dV}{dt} = \frac{dV}{dr} \cdot \frac{dr}{dt} = \frac{dV}{dr} \cdot \frac{dr}{dS} \cdot \frac{dS}{dt}$$

$$\text{where } \frac{dS}{dr} = 8\pi r \quad \text{and} \quad \frac{dV}{dr} = 4\pi r^2$$

$$\therefore \frac{dV}{dt} = 4\pi r^2 \cdot \frac{1}{8\pi r} \cdot k = \frac{k}{2} \cdot r \propto r$$

$$(c) (i) \text{ We need } |1-x| \leq 1$$

$$\Rightarrow -1 \leq x-1 \leq 1 \quad \text{i.e. domain is } 0 \leq x \leq 2$$



Question 5

(a) For $n=1$, $LS = 1^3 + 3 \times 1^5 = 4$
 $RS = \frac{1}{2} \cdot 2^3 = 4$

$\therefore$ Assume $(1^3 + 3 \cdot 1^5) + \dots + (n^3 + 3n^5) = \frac{1}{2} n^3 (n+1)^3$ for some integer $n \geq 1$

Then $(1^3 + 3 \cdot 1^5) + \dots + (n^3 + 3n^5) + ((n+1)^3 + 3(n+1)^5)$
 $= \frac{1}{2} n^3 (n+1)^3 + (n+1)^3 + 3(n+1)^5$, using the assumption

$$= \frac{1}{2} (n+1)^3 (n^3 + 2 + 6(n+1)^2)$$

$$= \frac{1}{2} (n+1)^3 (n^3 + 6n^2 + 12n + 8)$$

$$= \frac{1}{2} (n+1)^3 (n+2)^3 = RS \text{ for } n+1$$

$\therefore$ if correct for n it's correct for $n+1$

But, it is correct for $n=1$ $\therefore$ by induction it's correct

(b) (i) $\frac{C_k}{C_{k-1}} = \frac{\binom{n}{k}}{\binom{n}{k-1}} = \frac{n! (n-k+1)! (k-1)!}{(n-k)! k! n!} = \frac{n-k+1}{k}$

(ii) From (i), $\sum_{k=1}^n \frac{k C_k}{C_{k-1}} = \sum_{k=1}^n (n-k+1)$

$$= n + (n-1) + \dots + 2 + 1, \text{ arithmetic series}$$

$$= \frac{n}{2} (n+1)$$

$$(c) \quad (i) \quad f(-1) = -1 + A + 2 - A - 1 = 0$$

$\therefore x+1$ is a factor of $f(x) \quad \forall A$

(ii) From (i) and $f(x)$ we must have

$$f(x) = (x+1)^2(x-1)$$

$$\therefore f(1) = 1 + A + 2 + A - 1 = 0$$

$$\Rightarrow A = -1$$

(iii) From (i),

$$f(x) = (x+1)(x^2 - Ax - 1) \equiv x^3 + (A+2)x^2 + Ax - 1$$

$\therefore$ equating coefficients of x

$$-1 - A = A$$

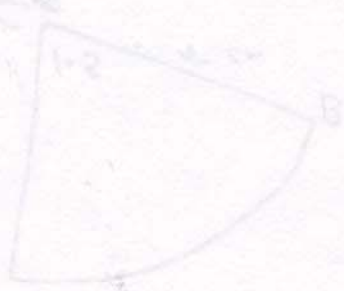
$$\therefore A = -\frac{1}{2}$$

[Alternatives abound]

$$\text{e.g. Sum of roots} = -1 + A = -(A+2)$$

$$\text{or put } x=1 \quad \dots \quad 2(-A) = A+2 + A$$

etc



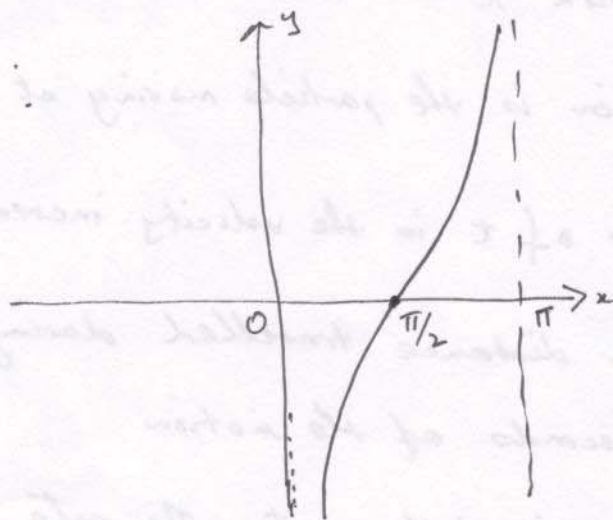
Question 6

$$(a) \quad (i) \quad f'(x) = \frac{1}{\tan \frac{x}{2}} \cdot \frac{1}{2} \sec^2 \frac{x}{2}$$

$$= \frac{1 + \tan^2 \frac{x}{2}}{2 \tan \frac{x}{2}} = \operatorname{cosec} x$$

$$\text{Since } \sin x = \frac{2t}{1+t^2} \quad \text{if } t = \tan \frac{x}{2}$$

$$(ii) \quad y=0 \Rightarrow \tan \frac{x}{2} = 1 \Rightarrow \frac{x}{2} = \frac{\pi}{4}, \quad x = \frac{\pi}{2}$$



[Note, $\operatorname{cosec} x > 0$ if $0 < x < \pi$]

$$(iii) \quad f^{-1}(x) : x = \ln \left(\tan \frac{y}{2} \right)$$

$$\therefore \tan \frac{y}{2} = e^x$$

$$\Rightarrow \frac{y}{2} = \tan^{-1}(e^x)$$

$$\therefore y = 2 \tan^{-1}(e^x)$$

$$(b) \quad (i) \quad \frac{1}{2} v^2 = \frac{1}{2} (20-x)^2$$

$$\therefore \quad \frac{d(\frac{1}{2} v^2)}{dx} = \frac{1}{2} \cdot 2(20-x)(-1) \\ = -(20-x) = -v$$

$$(ii) \quad \therefore \quad \frac{dv}{dt} = -v$$

$$\Rightarrow \quad \frac{dt}{dv} = -\frac{1}{v} \quad \text{or} \quad -\frac{dt}{dv} = \frac{1}{v}$$

$$\therefore \quad -t = \ln v + c \quad : \quad t=0, x=19, v=1$$

$$\therefore \quad 0 = \ln 1 + c, \quad c=0$$

$$\therefore \quad \ln v = -t$$

$$\Rightarrow \quad v = e^{-t}$$

Question 7

(a) (i) at P, $\dot{y} = 0$

$$\therefore -10T + V \sin d = 0 \quad \text{ie} \quad T = \frac{V \sin d}{10}$$

$$\begin{aligned} \text{(ii)} \quad \tan \beta &= \frac{PQ}{OQ} = \frac{-5T + V \sin d}{V \cos d} \\ &= \frac{-\frac{V \sin d}{2} + V \sin d}{V \cos d} \\ &= \frac{\sin d}{2 \cos d} = \frac{1}{2} \tan d \end{aligned}$$

$$\begin{aligned} \text{(iii)} \quad h &= -5 \left(\frac{V \sin d}{10} \right)^2 + V \sin d \left(\frac{V \sin d}{10} \right) \\ &= -\frac{V^2 \sin^2 d}{20} + \frac{V^2 \sin^2 d}{10} = \frac{V^2 \sin^2 d}{20} \end{aligned}$$

$$\therefore 20h = V^2 \sin^2 d$$

$$\text{(iv)} \quad \text{From (iii)} \quad \frac{V^2}{5h} = \frac{4}{\sin^2 d} = 4 \operatorname{cosec}^2 d$$

$$= 4(1 + \cot^2 d)$$

$$= 4 + (2 \cot d)^2$$

$$= 4 + \cot^2 \beta, \text{ from (ii)}$$

$$\therefore V^2 = 5h(4 + \cot^2 \beta)$$

(b) (i) From data, tangent at $P(2p, p^2)$ is

$$2px = 2(y + p^2)$$

$$\text{i.e. } px = y + p^2$$

(ii) Tangent at Q is $qx = y + q^2$

$$\therefore \text{ at } T, (p-q)x = p^2 - q^2 = (p-q)(p+q)$$

$$\Rightarrow x = p+q$$

$$\therefore y = p(p+q) - p^2 = pq$$

$$\text{i.e. } T = (p+q, pq)$$

(iii) PQ is the chord of contact from T

$$\therefore \text{ from data, } PQ \text{ is } (p+q)x = 2(y + pq)$$

(iv) $R(x_0, y_0)$ is on PQ

$$\therefore \text{ from (iii), } (p+q)x_0 = 2(y_0 + pq)$$

This last equation means that

$$T(p+q, pq) \text{ is on the line } x_0x = 2(y + y_0)$$

i.e. T lies on the line containing the chord of contact from R